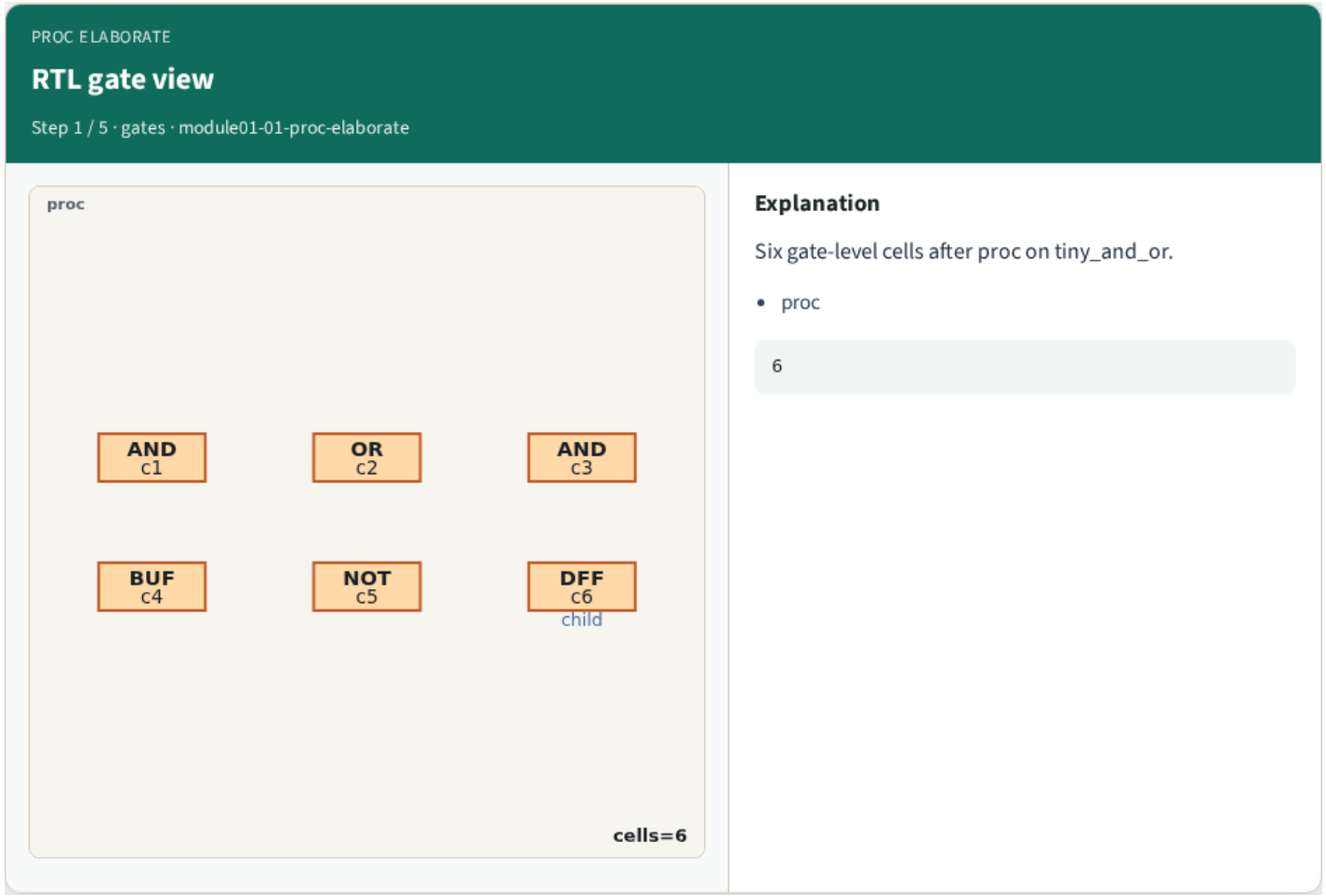
Why proc

The proc pass lowers RTL to a gate-level netlist Yosys can optimize

The idea

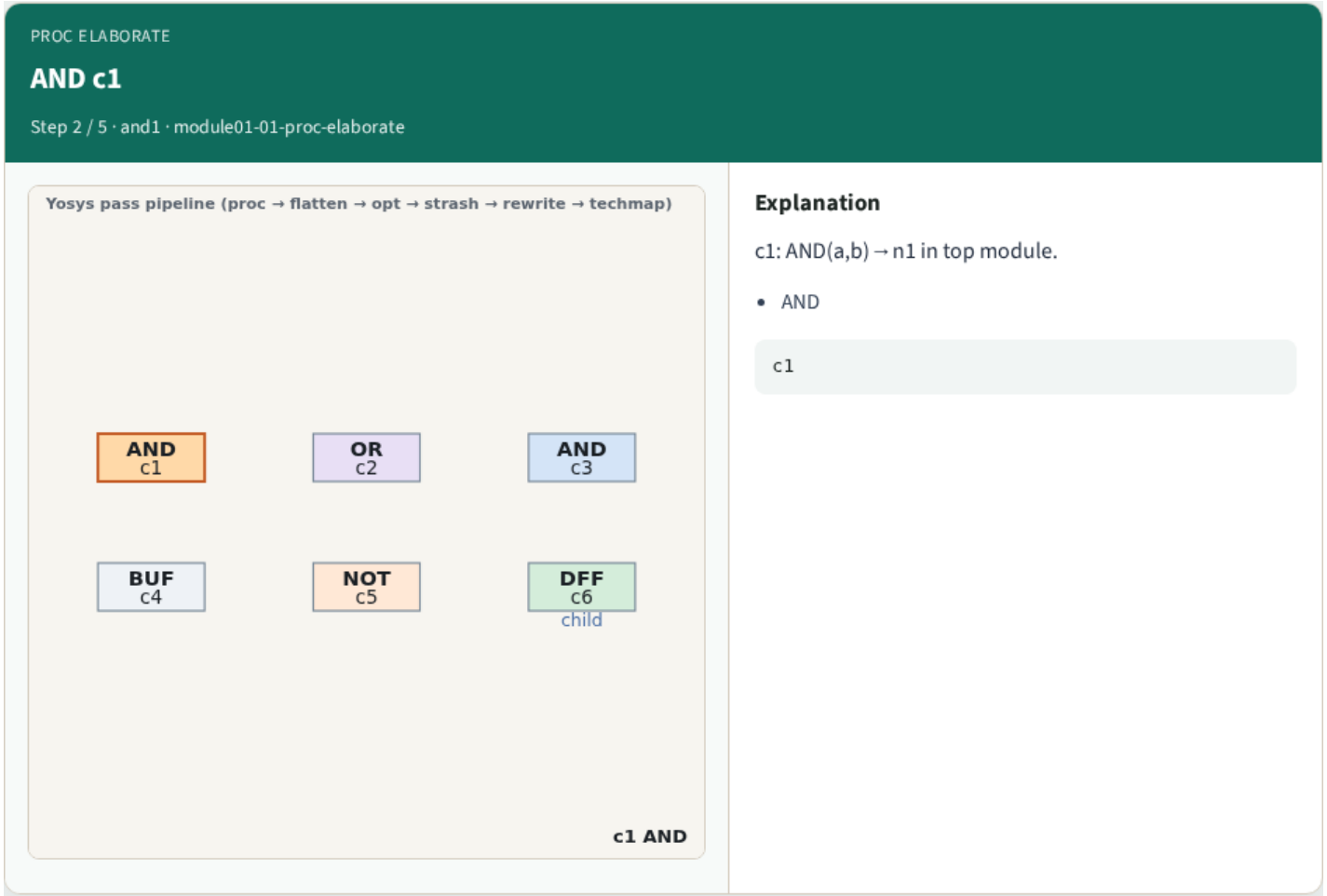
- GOLDEN: rtl_cell_count=6, module_count=2

RTL gate view



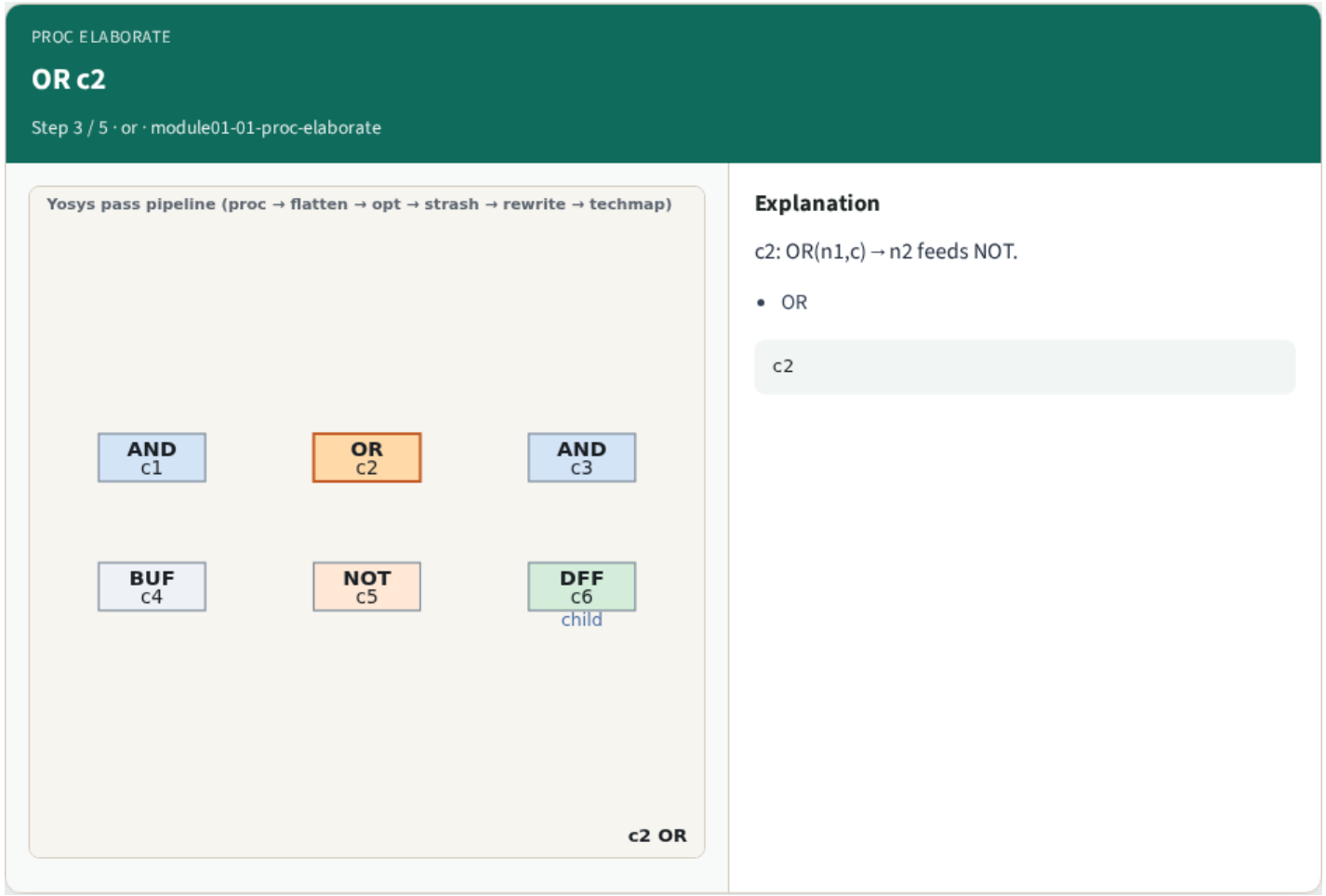
RTL gate view

AND c1


PROC ELABORATE
AND c1
Step 2 / 5 · and1 · module01-01-proc-elaborate
Yosys pass pipeline (proc → flatten → opt → strash → rewrite → techmap)
AND
c1
OR
c2
AND
c3
BUF
c4
NOT
c5
DFF
c6
child
c1 AND
Explanation
c1: AND(a,b) → n1 in top module.
AND
c1

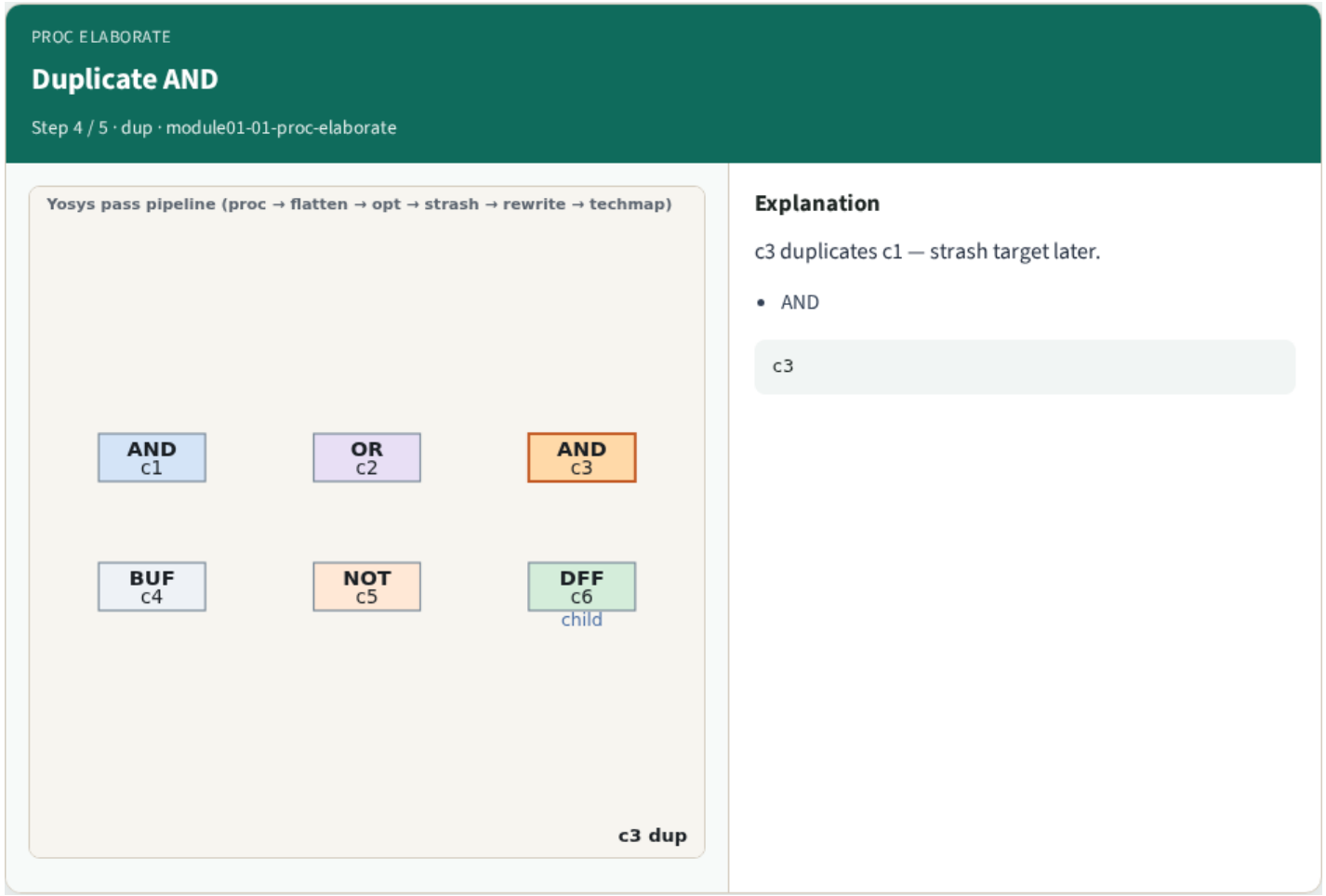
AND c1

OR c2


PROC ELABORATE
OR c2
Step 3 / 5 · or · module01-01-proc-elaborate
Yosys pass pipeline (proc → flatten → opt → strash → rewrite → techmap)
AND c1
OR c2
AND c3
BUF c4
NOT c5
DFF c6
child
c2 OR
Explanation
c2: OR(n1,c) → n2 feeds NOT.
OR
c2

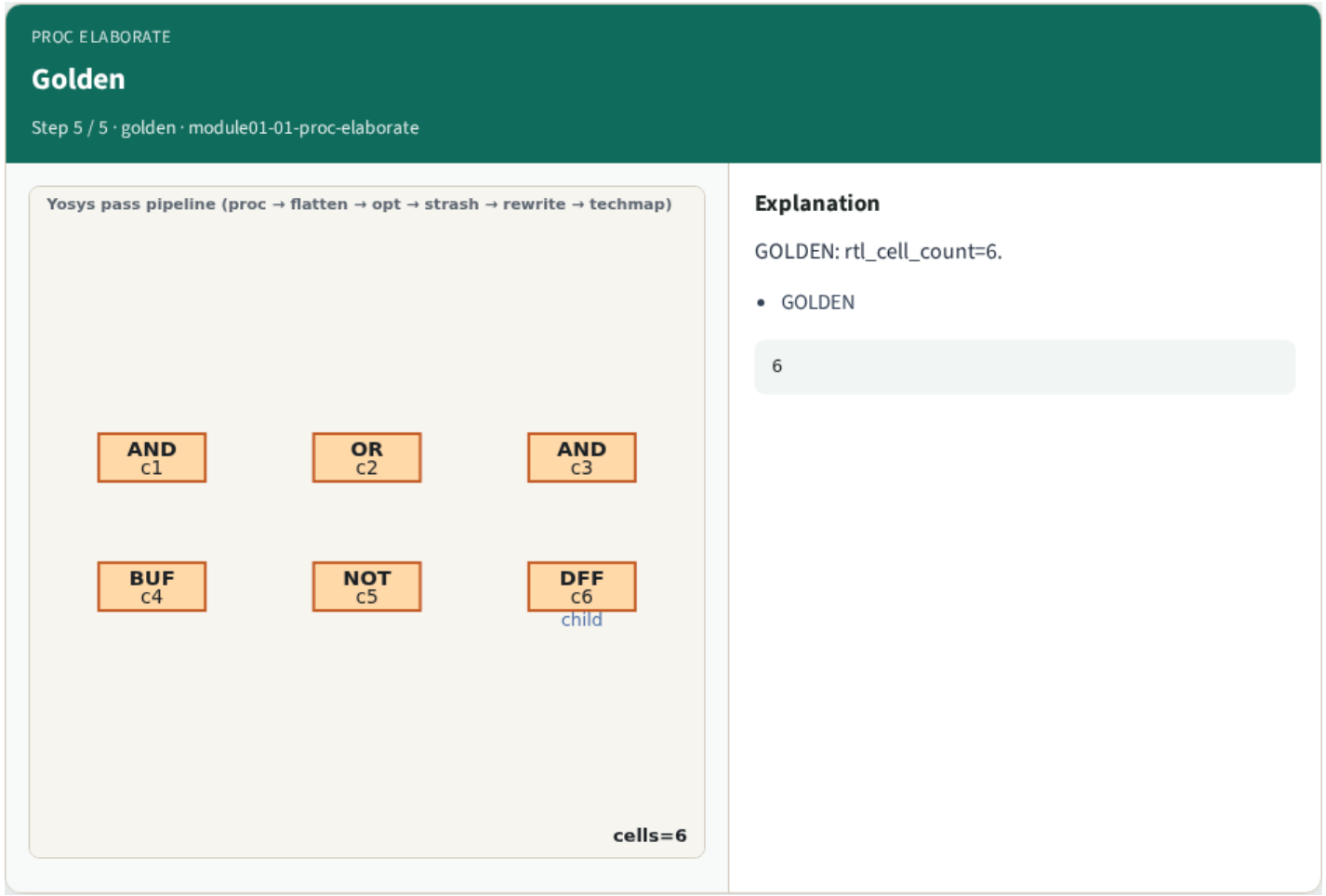
OR c2

Duplicate AND



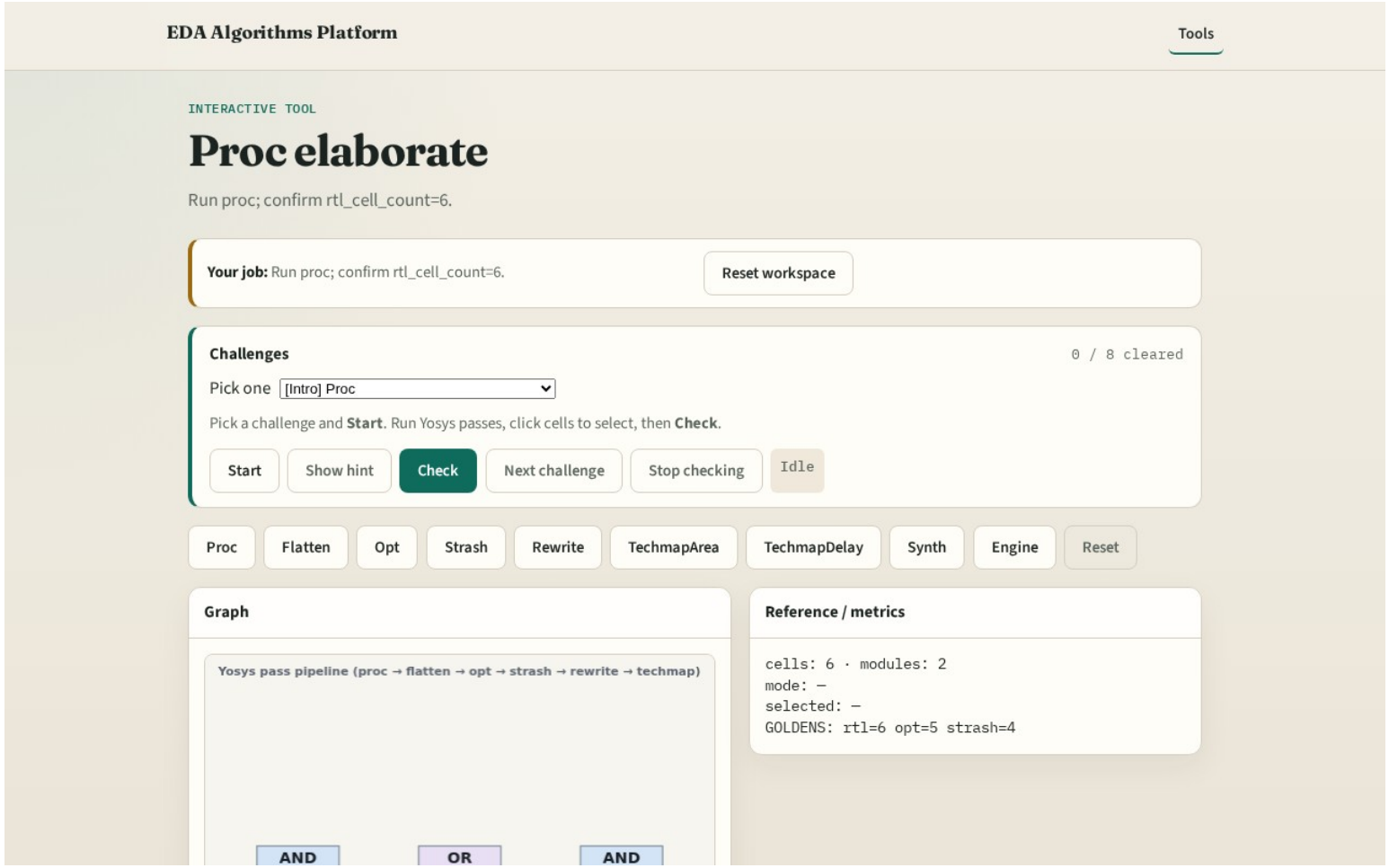
Duplicate AND

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call pass_proc(); assert cell_count==6

Pitfalls

- Confusing proc with synth
- Expecting fewer cells before opt

Your turn

- Next: hierarchy flatten